

The conjectured generating function of OEIS A121138, from the recurrence the entry states

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Abstract

OEIS A121138 states a linear recurrence as a fact and carries a rational generating function as an empirical conjecture. The conjecture follows from the recurrence by polynomial algebra alone: the recurrence fixes the denominator, the published terms fix the numerator exactly, and the generating function is then determined as a rational function rather than checked on finitely many coefficients. No model of the sequence and no reading of its name is required.

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1 What the entry says

The first terms are 1, 1, 3, 10, 46, 192, 840, 3584, $\dots$, with offset 1. The entry states, as a fact,

$$a(n) = 8*a(n-1) - 12*a(n-2) - 32*a(n-3) + 64*a(n-4) \text{ for } n > 7.$$

and separately records the conjecture

$$\text{Empirical G.f.: } -3/128 + x^3/4 + x^2/2 + 49x/64 - (296x^3 - 140x^2 + 6x + 3)/(128(2x-1)(2x+1)(4x-1)^2). \text{ - _Vaclav Kotesovec_ , Nov 29 2012}$$

The stated line was taken from the entry's factual formulas only. A formula qualified by a finite range – "holds at least up to $n = 1000$ but is not known to hold in general" – is not a fact and is not used, nor is anything inside a conjectural block.

2 The argument

Write $b_k = a(1+k)$ and

$$D(x) = 1 - \sum_i c_i x^i = -64x^4 + 32x^3 + 12x^2 - 8x + 1,$$

the polynomial of the stated recurrence, of degree 4. Let $B(x) = \sum_{k \geq 0} b_k x^k$.

The recurrence says $b_k = \sum_i c_i b_{k-i}$ for every $k \geq 7$. The coefficient of x^k in $D(x)B(x)$ is exactly $b_k - \sum_i c_i b_{k-i}$, so it vanishes for every $k \geq 7$ and

$$D(x)B(x) = P(x) = \sum_{k < 7} \left(b_k - \sum_i c_i b_{k-i} \right) x^k = -16x^7 - 24x^6 - 30x^5 + 30x^4 + 7x^3 - 7x^2 + x,$$

a polynomial whose every coefficient is fixed by finitely many published terms. Therefore

$$B(x) = \frac{P(x)}{D(x)} = \frac{x^3}{4} + \frac{x^2}{2} + \frac{49x}{64} - \frac{3}{128} - \frac{296x^3 - 140x^2 + 6x + 3}{(2x+1)(4x-1)^2(256x-128)},$$

which is the conjectured expression.

Theorem 1. *The conjectured generating function of OEIS A121138 is correct, given the recurrence the entry states.*

Proof. Immediate from the displayed identity: $DB = P$ holds coefficientwise, $D(0) = 1 \neq 0$ so D is invertible in the ring of formal power series, and $B = P/D$ follows. This is an identity of rational functions, not a comparison of the 26 terms the entry publishes – those enter only in fixing the 7 coefficients of P . $\square$